

Software Requirements Specification

for

NexusNotes

Project Team: Code Club

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Contents

1	Introduction	3
1.1	Purpose	3
1.2	Document Conventions	3
1.3	Intended Audience	3
1.4	Product Scope	3
1.5	References	4
2	Overall Description	4
2.1	Product Perspective	4
2.2	Product Functions	4
2.3	User Classes and Characteristics	4
2.4	Operating Environment	5
2.5	Design and Implementation Constraints	5
2.6	User Documentation	5
2.7	Assumptions and Dependencies	5
3	External Interface Requirements	5
3.1	User Interfaces	5
3.2	Hardware Interfaces	5
3.3	Software Interfaces	6
3.4	Communications Interfaces	6
4	System Features	6
4.1	Upload Academic Resource	6
4.2	Search and Download Resource	6
4.3	View Tech-Trend Feed	7
4.4	Rate and Review Resource	7
4.5	User Management	7
4.6	Content Moderation	8
5	Other Nonfunctional Requirements	8
5.1	Performance Requirements	8
5.2	Security and Privacy Constraints	8
5.3	Software Quality Attributes	9
6	Other Requirements	9
6.1	Applicable Standards	9

1 Introduction

1.1 Purpose

The purpose of this document is to define the high-level objectives, scope, and strategic direction of the Nexus Notes platform. It provides a comprehensive overview of the system's goals, target users, stakeholders, and core functionalities. This document serves as a foundational reference for developers, academic stakeholders, and institutional authorities by clearly outlining the problem domain and the proposed solution. It also ensures a shared understanding of the system vision among all participants involved in the project lifecycle.

1.2 Document Conventions

This document was created based on the IEEE template for System Requirement Specification Documents. The requirements are prioritized as High, Medium, or Low based on their critical impact on the system's core functionalities.

1.3 Intended Audience

- **Students:** Access, contribute, and utilize academic resources.
- **Faculty Members:** Verify and validate academic content.
- **Nexus Notes Administrators:** Manage platform operations, user roles, and content governance.
- **Developers:** Design, develop, and maintain the system.
- **Academic Coordinators:** Align content with curriculum requirements.

1.4 Product Scope

The fragmentation of academic resources across multiple platforms creates significant inefficiencies for students seeking reliable study materials, often resulting in redundant efforts and information silos. Nexus Notes is a web-based centralized academic resource management system designed for educational institutions to resolve this issue. The platform aims to unify fragmented academic resources into a structured digital repository organized subject-wise and module-wise. The proposed platform assigns dedicated digital repositories to every curriculum subject, segmented into specialized subsections for lecture notes, critical question banks, capstone project archives, and emerging technology trends. The system integrates dynamic collaborative features, including a contribution mechanism for peer-to-peer resource sharing, a faculty verification layer to ensure content accuracy, and a categorization logic that prioritizes high-quality content based on user engagement. Furthermore, to bridge the gap between theoretical coursework and industry demands, the system incorporates a real-time Tech-Trend feed that aggregates relevant technological advancements for each specific domain. The scope of the current system does not include full-fledged learning management functionalities such as automated grading systems, examination modules, or institutional administrative processes.

1.5 References

- Vision Document, Nexus Notes, Centralized Academic Resource Platform, Version 1.0, January 30, 2026.
- Use Cases for Nexus Notes, Version 1.0, February 13, 2026.

2 Overall Description

2.1 Product Perspective

Nexus Notes is designed as a centralized academic knowledge platform that integrates students, faculty, and academic resources into a unified digital ecosystem. The system acts as a bridge between fragmented academic materials and structured learning processes. It complements existing institutional systems by providing an organized repository for academic content, collaboration features, and real-time technology updates. Nexus Notes operates as a web-based platform accessible to all authorized users within an academic institution.

2.2 Product Functions

The core capabilities of Nexus Notes include:

- **Centralized Repository:** Provides subject-wise digital repositories for academic resources.
- **Collaborative Sharing:** Enables students and faculty to contribute and share content.
- **Content Verification:** Allows faculty members to validate and approve academic materials.
- **Intelligent Categorization:** Organizes content based on relevance and usage.
- **Advanced Search:** Supports fast and accurate retrieval of academic resources.
- **Tech-Trend Integration:** Displays real-time technology trends related to academic domains.

2.3 User Classes and Characteristics

- **Primary Users:** Students who access and share academic resources. Students require easy access to reliable and structured academic resources to enhance learning efficiency.
- **Secondary Users:** Faculty members who review and verify content. Faculty members focus on ensuring content accuracy and academic relevance.
- **Administrative Users:** Nexus Notes administrators responsible for managing the platform, moderating content, and controlling user permissions.
- **Contributors:** Users who upload and curate academic content.

2.4 Operating Environment

Users interact with Nexus Notes through web browsers on devices such as laptops, tablets, and smartphones. The platform is accessed in various environments including classrooms, libraries, hostels, homes, and laboratories. The system supports collaborative academic activities with continuous updates of digital content.

2.5 Design and Implementation Constraints

The Nexus Notes platform is subject to technical limitations such as server capacity, storage availability, network bandwidth, and system scalability. The performance of the system depends on reliable internet connectivity and compatible hardware devices. The system must ensure data security, user privacy, and protection of academic content. Compliance with institutional data policies and ethical standards is required.

2.6 User Documentation

- **User Manual:** A user manual is provided to guide students, faculty, and administrators in using the Nexus Notes platform.
- **Administrator Guide:** An administrator guide describes system configuration, user management, and content moderation procedures.
- **Developer Documentation:** Technical documentation supports system maintenance, future enhancements, and integration with other institutional systems.

2.7 Assumptions and Dependencies

It is assumed that users have access to stable internet connectivity and compatible digital devices. The platform depends on institutional support for adoption and faculty participation for content verification. Additionally, the effectiveness of the system relies on active user engagement and continuous content contribution.

3 External Interface Requirements

3.1 User Interfaces

Users interact with Nexus Notes through modern web browsers. The interface is divided into role-based dashboards:

- **Student Dashboard**
- **Administrator Console**

3.2 Hardware Interfaces

The platform supports cross-platform access via modern web browsers.

3.3 Software Interfaces

The platform requires a web server, database management system, and stable internet connectivity. It may integrate with institutional authentication systems and external tech APIs.

3.4 Communications Interfaces

Nexus Notes requires an active internet connection to function.

4 System Features

4.1 Upload Academic Resource

Description: A student uploads academic materials such as lecture notes, project reports, or question papers to a subject repository for sharing.

Priority: High

Main Flow:

- Student accesses the “Contribute” section.
- Selects category and uploads file (PDF, DOCX, PPT).
- Adds required metadata (Subject, Topic, Tags).
- System validates and publishes the resource.

Key Requirements:

- Student must be logged in with Contributor permissions.
- Maximum file size is 50MB.
- Resource is stored and indexed in the database.

4.2 Search and Download Resource

Description: Students search academic materials using keywords or filters and download them for study.

Priority: Medium

Main Flow:

- Student enters keywords or applies filters.
- System displays ranked results.
- Student downloads selected resource.

Key Requirements:

- Student must be logged in.
- Download count is updated.

4.3 View Tech-Trend Feed

Description: Students view real-time technological updates related to their academic domains.

Priority: Medium

Main Flow:

- Student accesses the Tech-Trend section.
- System fetches and displays curated updates.

Requirement:

- Active internet connection is required.

4.4 Rate and Review Resource

Description: Students provide ratings and optional reviews for shared resources.

Priority: Low

Main Flow:

- Student selects rating (1–5 stars).
- Optional comment is submitted.
- System updates the average rating.

Requirement:

- Student must have accessed the resource.

4.5 User Management

Description: Nexus Notes administrators manage user roles and permissions.

Priority: High

Main Flow:

- Administrator accesses the User Management section.
- Updates user roles or permissions.
- System saves changes.

Requirement:

- Administrator must be logged in with appropriate privileges.

4.6 Content Moderation

Description: Nexus Notes administrators review and manage uploaded content.

Priority: High

Main Flow:

- Administrator reviews flagged resources.
- Deletes or approves content.
- System updates content visibility.

Requirement:

- Content must be flagged or under review.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

The Nexus Notes platform shall operate efficiently under normal and peak usage conditions.

- The system shall support concurrent access by multiple users without noticeable performance degradation.
- Search results shall be displayed within 2–3 seconds under normal load.
- File uploads up to 50MB shall be processed and validated efficiently.
- The platform shall be scalable to accommodate increasing numbers of users and resources.
- System availability shall remain high except during scheduled maintenance.

5.2 Security and Privacy Constraints

The system shall ensure protection of user data and academic resources.

- User authentication shall be required to access system features.
- Role-based access control shall restrict actions based on user type (Student, Faculty, Administrator).
- Passwords and sensitive data shall be securely stored.
- Secure communication protocols shall be used for data transmission.
- The system shall comply with institutional data protection policies.

5.3 Software Quality Attributes

Nexus Notes shall follow standard software development practices to ensure reliability and maintainability.

- The system shall provide a user-friendly and intuitive interface.
- The platform shall support responsive design for different devices.
- Periodic backups shall be maintained to prevent data loss.

6 Other Requirements

6.1 Applicable Standards

Database updates and backups ensure academic resources are securely managed. The system provides a mechanism for continuous updates of digital content.